

Dietary modification of human macular pigment density.

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PURPOSE: The retinal carotenoids lutein (L) and zeaxanthin (Z) that form the macular pigment (MP) may help to prevent neovascular age-related macular degeneration. The purpose of this study was to determine whether MP density in the retina could be raised by increasing dietary intake of L and Z from foods. **METHODS:** Macular pigment was measured psychophysically for 13 subjects. Serum concentrations of L, Z, and beta-carotene were measured by high-performance liquid chromatography. Eleven subjects modified their usual daily diets by adding 60 g of spinach (10.8 mg L, 0.3 mg Z, 5 mg beta-carotene) and ten also added 150 g of corn (0.3 mg Z, 0.4 mg L); two other subjects were given only corn. Dietary modification lasted up to 15 weeks. **RESULTS:** For the subjects fed spinach or spinach and corn, three types of responses to dietary modification were identified: Eight "retinal responders" had increases in serum L (mean, 33%; SD, 22%) and in MP density (mean, 19%; SD, 11%); two "retinal nonresponders" showed substantial increases in serum L (mean, 31%) but not in MP density (mean, -11%); one "serum and retinal nonresponder" showed no changes in serum L, Z, or beta-carotene and no change in MP density. For the two subjects given only corn, serum L changed little (+11%, -6%), but in one subject serum Z increased (70%) and MP density increased (25%). **CONCLUSIONS:** Increases in MP density were obtained within 4 weeks of dietary modification for most, but not all, subjects. When MP density increased with dietary modification, it remained elevated for at least several months after resuming an unmodified diet. Augmentation of MP for both experimental and clinical investigation appears to be feasible for many persons.